# THEODORE E. PADILLA

GIS Specialist | Geologist
Quezon City, Metro Manila, Philippines 1101

## Experience

**Philippine Institute of Volcanology and Seismology (DOST-PHIVOLCS)** Quezon City, Metro Manila

***Junior GIS Specialist*** August 2025 – Present

*GATES Project 3: Upscaling and Integrating Geospatial Analytics and AI in Planning and Decision Making*

- Developed Python-based GIS automation scripts to integrate multi-agency GIS models, streamlining the process of data dissemination, map layout, and report generation through a centralized web platform
- Developed service area analysis, suitability analysis and risk calculation Python modules to support research and disaster planning and decision making
- Performed comparative analysis of Python and GIS-based models and created corresponding map layouts to monitor model improvements and support the presentation of agency use cases
- Conducted consultations for national government agencies through focus group discussions to provide geospatial solutions
- Won 3rd place in a hackathon leveraging AWS and AI tools to prototype an interactive, AI-powered geospatial application for historical archives

***Science Research Specialist I*** March 2024 – August 2025

*GeoRiskPH and Innovations Laboratory (InnoLab) Project*

- Cleaned and managed baseline exposure and topological data, ensuring accurate visualization and analysis for local database managers and supporting disaster planning through the PlanSmart application
- Conducted suitability analysis and assessment for earthquake evacuation open spaces in Iloilo City, aiding the local disaster unit in designating appropriate sites for earthquake evacuation
- Created ArcGIS web maps, applications and dashboards to support exposure data mapping, analysis and visualization
- Facilitated trainings and lectures on exposure database mapping (GeoMapperPH), natural hazards, and disaster preparedness for national and local government units in the Philippines
- Authored a research poster on the suitability assessment of safe open spaces for earthquakes in Iloilo City presented in GeoCon 2024

***Science Research Assistant*** November 2023 – March 2024

*Geospatial Information Management and Analysis (GIMA) Project*

- Produced automated map templates of hazard map layouts, reducing production time for provincial and municipal-scale map layouts by 50 percent
- Performed change detection analysis of landslides induced by the 2023 Offshore Davao Occidental Earthquake, serving as baseline data for field validation and for a research poster presented in GeoCon 2023
- Generated ash dispersion and trajectory models to support the monitoring of Mayon Volcano during its activity in 2023

**Teragrail Geology and Geotechnics** Imus City, Cavite

***Junior Geologist*** May 2023 – October 2023

- Supervised borehole drilling operations for a solar power project in Batangas, ensuring accurate standard penetration test (SPT) data
- Processed LiDAR and optical images in Agisoft Metashape to produce topographic contours, aiding the geodetic engineer in drafting a topographic plan
- Consolidated land parcels from titles using AutoCAD and drafted consolidation plans, maintaining data integrity by cross-validating plotted areas against land title descriptions
- Drafted Engineering, Geological and Geohazard Assessment Reports (EGGAR) with supporting geologic and geohazard maps created in ArcGIS and QGIS, facilitating regulatory approval for infrastructure projects

## Education

**Mapúa University – Intramuros** City of Manila, Metro Manila

***Double Degree in Bachelor of Science in Geology and Geological Science and Engineering*** July 2015 – July 2022

- Top 10 Academic Performer of Bachelor of Science in Geological Science and Engineering – Batch 2015 (2016)
- Member, Association of Geologists and Geological Engineers of Mapúa

## Licenses

**Geologist**

***Professional Regulation Commission***

February 2023 – June 2029

## Research Works

**Suitability Assessment of GIS-generated Safe Open Spaces for Earthquakes in Iloilo City: A Case Study**
Mercado, N.J.L., Tatlonghari, C.R.A., Gomez, G.V.V.T., Padilla, T.E., Palma, J.A., Almoneda, K.C.A., Tabuzo, J.H.B., Cahulogan, M.T.
Research Poster – GeoCon 2024

- Conducted suitability analysis of GIS-automated temporary earthquake evacuation open spaces using field-assessed weighted parameters such as accessibility, ownership, current and future use, vicinity access and pedestrian road access

**Shake to Slide: Analysis of the Landslides Triggered by the 17 November 2023 Mw 6.8 Offshore Davao Occidental Earthquake**
Fermo, D.G.S., Daag, A.S., Dizon, M.P., De Leon, R.J.B., Pagbilao, V.T.M., Dipatuan, J.M., Baltazar, H.M.M., Buhay, D.J.L., Pidlaoan, A.C., Padilla, T.E.
Research Poster – GeoCon 2024

- Performed preliminary change detection analysis of earthquake-induced landslides from the 17 November 2023 Magnitude 6.8 Offshore Davao Occidental Earthquake using PlanetScope true color and false color composite imagery, used as baseline data for field validation

**Flood Risk Assessment and Dam Breach Analysis of Magat Dam Using HEC-RAS and LU/LC Data: A Multiple Criteria Analysis**
Basan, A.H.G., Padilla, T.E., De la Peña, F.
Undergraduate Thesis – 2022

- Modeled dam breach scenarios for Magat Dam in HEC-RAS using Typhoon Ulysses precipitation data; utilized geospatial overlay analysis to compare overtopping and piping failure scenarios, identifying agricultural crops as the most vulnerable land cover category

## Relevant Skills

**Technical Skills**

- GIS – used various geoprocessing and layout tools in ArcGIS Pro, ArcMap and QGIS, notably map series, topology checker, ModelBuilder, suitability analysis, and road network analysis
- Python programming – used GeoPandas, Shapely, Rasterio, GDAL, NetworkX and ArcPy libraries in Python-based GIS workflows
- Remote Sensing – applied land cover change detection in ArcGIS for landslides, and image classification in Google Earth Engine for remote validation of open spaces
- Photogrammetry – generated contours, digital elevation and terrain model, and orthomosaic in Agisoft Metashape
- Database management – practiced data cleansing, pivot tables and automation in Excel and ArcGIS in managing national to barangay-level baseline data
- HEC-RAS and HEC-HMS – utilized dam breach modelling and flood modelling for undergraduate thesis
- AutoCAD – drafted parcel consolidation plans and topographic plans
- AI and Cloud Integration – completed in-house modules on AWS Cloud Fundamentals and AI tools, RAG frameworks, and Podman Containerization
- Technical Writing – wrote technical reports, web articles, research poster, technical presentation

**Soft Skills**

- Technical communication – lectured on earthquake and volcanic hazards, database management, QGIS and ArcGIS data conversion and projection for national and local government units
- Mentoring and training – trained government staff and trainers on exposure data mapping and database management
- Leadership and management – supervised geotechnical drilling operations
- Coordination and facilitation – coordinated with government units and private clients for fieldwork and training cooperation

## References

**Gilson Andre Narciso**
Geospatial Python Developer
Datafarming, Inc.
(Contact details provided upon request)

**Shania Bagaforo**
GIS Analyst
MasTec Communications Group
(Contact details provided upon request)